# Representation Resolution Module (RRM)

**OOF™ Origin Open Foundation™**

Independent Methodological Authority

**Architecture Ecosystem:** Structured Reality Standards™
**Architecture Family:** SIMULOS® — Simulation Governance Architecture
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## Governed Space

## Representation Resolution

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## Canonical Definition

Representation Resolution Module (RRM) defines the governance conditions under which the granularity, discretization, segmentation, temporal interval, spatial scale, state detail, interaction detail, and representational precision of a simulation are established, justified, assessed, bounded, and maintained relative to the governed Simulation Purpose, Scope, Relevant Reality, approved abstractions, required fidelity, scenarios, and intended interpretation.

RRM ensures that simulation representation operates at a level of resolution capable of exposing the distinctions, transitions, interactions, variations, and events that materially matter to the simulation inquiry without requiring unnecessary representational detail that does not materially improve the simulation outcome.

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## Operational Role

The preceding modules establish:

RSM — What parts of reality must the simulation represent?

RAM — How may the complexity of that selected reality legitimately be reduced?

RFM — How faithfully must the resulting representation correspond to material reality?

RRM asks:

At what level of granularity must that reality be represented so that material simulation phenomena remain observable and distinguishable?

The governed progression is:

Reality Selection → Reality Abstraction → Representation Fidelity → Resolution Requirement → Resolution Design → Resolution Assessment → Resolution Gap → Resolution Status → Representation Limitation

RRM does not determine which reality belongs in the simulation.

It does not govern whether abstraction is legitimate.

It does not establish general fidelity.

It governs specifically whether the simulation representation is sufficiently granular to support the simulation inquiry.

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## Module Operational Space

RRM governs: resolution objectives, resolution requirements, resolution dimensions, resolution baselines, resolution thresholds, spatial resolution, temporal resolution, state resolution, behavioral resolution, interaction resolution, environmental resolution, population resolution, event resolution, process resolution, dependency resolution, failure resolution, measurement resolution, input resolution dependencies, output resolution dependencies, multi-resolution representation, adaptive resolution, cross-component resolution, resolution consistency, resolution sufficiency, resolution gaps, resolution trade-offs, resolution uncertainty, resolution sensitivity, resolution limitations, conditional resolution, resolution status, resolution change, and resolution traceability.

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## Module Function

A simulation may represent the correct parts of reality with acceptable fidelity and still fail because its representation is too coarse to expose material phenomena.

Examples include: a time step too large to capture a rapid failure, a spatial grid too coarse to represent a local hazard, a population represented only through averages when subgroup differences matter, operational states collapsed into one generic state, interactions aggregated so strongly that critical dependencies disappear, rare events removed by coarse event representation, or environmental variation represented at a scale that hides material local conditions.

RRM exists to prevent: coarse representation being mistaken for sufficient representation, unnecessary detail being mistaken for methodological strength, resolution being selected only according to available computational capacity, resolution being increased after observing desired results without governance, one high-resolution component masking materially coarse dependencies, material events disappearing between simulation intervals, averages hiding important distributions or subgroups, resolution changes silently altering simulation meaning, and simulation outputs implying greater precision than the underlying representation can support.

## Resolution Principle

SIMULOS® establishes:

Representation resolution is sufficient only when the simulation can distinguish the material phenomena, states, transitions, interactions, variations, and events required by the simulation inquiry.

Resolution therefore does not mean:

maximum detail.

It means:

sufficient granularity where granularity materially affects the simulation.

## Resolution Is Purpose-Relative

There is no universally correct simulation resolution.

Required resolution depends upon:

Simulation Purpose + Simulation Scope + Relevant Reality + Material Phenomena + Required Fidelity + Scenario + Intended Interpretation

A simulation designed to estimate:

annual regional energy demand

may legitimately use a substantially different temporal and spatial resolution from a simulation designed to govern:

millisecond-level electrical grid instability.

The correct resolution is therefore determined by the question being simulated, not by the maximum detail technically available.

## Resolution Object

A Resolution Object is any represented element, process, state, interaction, environment, population, event, dependency, or dimension whose level of granularity may materially affect simulation behavior or interpretation.

A Resolution Object may include: physical space, time, individual actors, populations, system states, environmental variables, interactions, dependencies, processes, failure mechanisms, events, or simulation components.

## Resolution Requirement

A Resolution Requirement establishes the minimum granularity necessary for a material Resolution Object.

It should derive from:

## Simulation Purpose & Scope

↓

## Relevant Reality

↓

## Material Phenomenon

↓

## Required Fidelity

↓

## Required Resolution

The resolution requirement should exist independently of whether the current simulation platform can technically achieve it.

## Resolution Dimensions

Resolution should not automatically be represented as one global characteristic.

A simulation may simultaneously possess:

## High Temporal Resolution

and:

## Low Spatial Resolution

or:

## High Component Resolution

and:

Low Interaction Resolution.

RRM therefore governs resolution dimensionally.

## Spatial Resolution

Spatial Resolution governs the granularity at which spatial reality is represented.

It may concern: grid size, geometric segmentation, distance intervals, terrain detail, object dimensions, location precision, infrastructure segmentation, environmental zones, or spatial boundaries.

The required spatial resolution depends upon whether smaller spatial differences can materially affect the simulation inquiry.

## Temporal Resolution

Temporal Resolution governs the granularity at which change over time is represented.

It may concern: simulation time steps, observation intervals, event intervals, update frequency, response timing, synchronization intervals, degradation periods, or temporal aggregation.

A temporal interval that is too large may cause a simulation to miss: rapid transitions, short-duration hazards, feedback effects, transient instability, collisions, cascading failures, or timing-sensitive interactions.

## State Resolution

State Resolution governs how finely operational or representational states are distinguished.

For example:

Operational

may be insufficient where the simulation inquiry requires distinction between:

Normal → Degraded → Restricted → Critical → Failed → Recovery.

Where different states produce materially different behavior, collapsing them may create a Resolution Gap.

## Behavioral Resolution

Behavioral Resolution governs the level at which differences in actor or system behavior must remain distinguishable.

A population represented by one average behavioral profile may be insufficient where material behavioral variation exists.

## Interaction Resolution

Interaction Resolution governs how finely interactions between simulation elements are represented.

This may include: frequency, sequence, direction, intensity, duration, conditionality, feedback, synchronization, and interaction state.

High component resolution cannot compensate automatically for insufficient interaction resolution.

## Environmental Resolution

Environmental Resolution governs the granularity at which environmental conditions are represented.

A single environment-wide value may be insufficient where local variation materially affects outcomes.

Relevant dimensions may include: temperature, weather, terrain, visibility, infrastructure, traffic, radiation, pressure, resource availability, or environmental hazards.

## Population Resolution

Population Resolution governs whether a population may legitimately be represented: collectively, by subgroup, by representative profiles, by distributions, or at individual-agent level.

The appropriate level depends upon whether differences between population members materially affect the simulation inquiry.

## Event Resolution

Event Resolution governs whether materially different events remain distinguishable.

Events should not be aggregated where their: sequence, timing, severity, cause, duration, location, or interaction

can materially change simulation outcomes.

## Process Resolution

Process Resolution governs the granularity at which a simulated process is decomposed.

A process represented as:

## Input → Output

may be sufficient for one simulation.

Another inquiry may require:

Input → Transformation A → Decision → Transformation B → Verification → Output.

RRM does not prescribe universal decomposition.

It governs whether the selected decomposition is sufficient.

## Dependency Resolution

Dependency Resolution governs whether materially different dependencies remain separately observable.

Multiple dependencies should not be merged into a generic relationship where their independent: failure, delay, strength, direction, or conditionality

may materially affect simulation behavior.

## Failure Resolution

Failure Resolution governs the granularity at which failures, degradation, and abnormal states must be represented.

A generic:

Failed

state may be insufficient where different failure modes create materially different consequences.

Failure resolution may therefore distinguish: partial degradation, sensor failure, communication failure, power failure, mechanical failure, software failure, dependency failure, cascading failure, or recovery failure.

## Measurement Resolution

Where simulation representation depends upon measured values, the resolution of those measurements may constrain legitimate simulation resolution.

A simulation cannot automatically create meaningful real-world precision merely by numerically subdividing low-resolution source information.

## Input Resolution Dependency

Simulation resolution may depend upon the granularity of input data.

Where required simulation resolution exceeds the meaningful resolution of available input information, the mismatch must remain visible.

Interpolation or synthetic expansion does not automatically resolve this mismatch.

## Output Resolution Dependency

Simulation outputs should not imply greater meaningful precision than the representation supporting them.

SIMULOS® establishes:

Output precision must not be mistaken for representation resolution.

A simulation may calculate twelve decimal places while the underlying representation supports only coarse distinctions.

## Resolution Baseline

A Resolution Baseline identifies the reference against which resolution sufficiency is assessed.

It may derive from: material event duration, physical scale, operational transition speed, environmental variation, population heterogeneity, interaction frequency, failure propagation, measurement capability, or another justified simulation requirement.

## Resolution Threshold

A Resolution Threshold defines the minimum acceptable granularity for a particular Resolution Object or dimension.

RRM does not prescribe universal thresholds.

Thresholds must be justified relative to material simulation phenomena.

## Resolution Assessment

A Resolution Assessment asks:

Can the current representation distinguish everything that materially needs to be distinguished for this simulation inquiry?

The assessment should consider not only nominal behavior but also relevant: transitions, boundaries, interactions, extremes, failures, rare events, and dynamic conditions.

## Resolution Sufficiency

Resolution is sufficient where further granularity would not materially change the simulation's ability to address the governed inquiry within its intended interpretation.

This does not mean that a finer representation could not exist.

It means the current resolution is methodologically sufficient for the purpose.

## Resolution Gap

A Resolution Gap exists where the current representation is too coarse to preserve a materially relevant distinction.

Examples include:

Temporal Gap — an event occurs between simulation steps.

Spatial Gap — a local phenomenon disappears inside a larger spatial unit.

Population Gap — materially different subgroups are merged.

State Gap — distinct operational states are collapsed.

Interaction Gap — important interaction differences are aggregated.

Failure Gap — materially different failure modes are treated as one.

## Resolution Gap Classification

A Resolution Gap may be classified as:

## Non-Material

## Material but Acceptable

Conditional

## Material and Unresolved

or:

## Resolution Undetermined

The classification must remain linked to the affected Simulation Purpose, Scope, scenario, and interpretation.

## Resolution Limitation

A Resolution Limitation identifies where insufficient resolution restricts legitimate simulation use or interpretation.

A limitation may apply to: particular scenarios, geographic areas, time intervals, populations, operational states, environmental conditions, failure modes, interactions, or outputs.

## Resolution Trade-Off

Resolution frequently creates trade-offs involving: computational demand, execution time, memory, data requirements, model complexity, scenario volume, repeatability, interpretability, and operational cost.

RRM permits these trade-offs.

It requires them to remain visible where they materially affect the simulation inquiry.

## Computational Constraint Rule

SIMULOS® establishes:

Available computational capacity may constrain achievable resolution, but it does not redefine the resolution required by the simulation purpose.

If required resolution cannot be achieved, the result should become a governed limitation rather than an undisclosed methodological compromise.

## Maximum Resolution Is Not the Objective

Increasing resolution can: increase computational cost, increase noise, amplify weak input assumptions, create false precision, reduce scenario coverage, complicate interpretation, and increase execution burden.

RRM therefore does not require maximum resolution.

## It requires justified resolution. Resolution ≠ Fidelity

Resolution and fidelity are distinct.

Resolution asks:

How finely is reality represented?

Fidelity asks:

How faithfully does that representation correspond to reality?

A simulation can be:

## High Resolution + Low Fidelity

or:

Lower Resolution + Sufficient Fidelity.

RFM governs fidelity.

## RRM governs resolution. Resolution ≠ Precision

Numerical precision does not establish representational resolution.

A coarse simulation represented with many decimal places remains coarse.

RRM therefore prevents numerical precision from being used as evidence of sufficient resolution.

## Multi-Resolution Representation

A simulation does not need identical resolution everywhere.

Different simulation components may legitimately operate at different resolutions.

For example: a critical subsystem may require millisecond resolution, surrounding infrastructure may require second-level resolution, long-term environmental change may require hourly or daily resolution.

The differences must remain methodologically justified and operationally compatible.

## Adaptive Resolution

Some simulations may change resolution dynamically.

Resolution may increase around: critical events, boundaries, high-risk states, failures, rapid transitions, or areas of uncertainty.

Adaptive resolution is permissible where the triggering logic and resulting representational effects remain governed.

## Adaptive Resolution Trigger

A resolution change should have an identifiable basis.

Possible triggers include: approaching threshold, rapid state change, detected anomaly, high uncertainty, emerging interaction, critical event, or scenario requirement.

## Cross-Component Resolution

Where multiple simulation components interact, their resolutions must be sufficiently compatible for the interaction being modeled.

A fast-changing component connected to a slowly updated dependency may create synchronization errors or hide material events.

## Resolution Compatibility

Resolution compatibility does not require identical resolution.

It requires that differences do not invalidate material interactions or interpretations.

## Resolution Synchronization

Where components operate at different temporal or spatial resolutions, the simulation should govern how information moves between them.

This may involve: aggregation, interpolation, synchronization, transformation, event bridging, or another governed mechanism.

## Resolution Transformation

When information moves between resolution levels, transformation may create information loss.

RRM requires material transformation effects to remain visible.

## Aggregation Loss

Aggregation may hide: extremes, rare events, local failures, subgroup differences, temporal peaks, nonlinear behavior, or causal relationships.

Where these phenomena matter, aggregation loss becomes a governance concern.

## Interpolation Risk

Interpolation can create values between observations.

Those values may be computationally useful but should not automatically be treated as independently observed reality.

Where interpolation materially influences simulation behavior, its role should remain traceable.

## Resolution Sensitivity

A Resolution Sensitivity Assessment asks:

Would materially plausible changes in representation resolution change the simulation outcome or interpretation?

Where results change substantially under reasonable resolution changes, the simulation may require stronger resolution governance.

## Resolution Convergence

Where appropriate, simulations may compare outcomes across increasingly fine resolutions.

If outputs stabilize sufficiently as resolution increases, this may support resolution sufficiency.

RRM does not require convergence testing universally.

It recognizes it as one possible basis for demonstrating that the chosen resolution is adequate.

## Resolution Uncertainty

Where the appropriate resolution cannot be confidently established, the uncertainty should remain explicit.

Resolution uncertainty may arise from: unknown event scales, limited real-world measurements, emerging phenomena, insufficient input granularity, uncertain failure behavior, or poorly understood interactions.

## Conditional Resolution

A representation may be sufficiently resolved only under defined conditions.

For example:

## Resolution Sufficient — Normal Operation

but:

Resolution Insufficient — Rapid Transient Failure.

This distinction must remain attached to simulation interpretation.

## Representation Resolution Status

RRM may establish:

## Resolution Sufficient

## Resolution Conditionally Sufficient

## Resolution Partially Sufficient

## Resolution Insufficient

or:

## Resolution Undetermined

The status should remain bound to the relevant Resolution Object, dimension, scope, scenario, and conditions.

## Weakest Material Resolution Principle

Where multiple resolution dimensions materially affect one simulation claim:

The claim cannot assume greater representational resolution than its materially limiting resolution dimension supports.

A high-resolution subsystem cannot erase a critical low-resolution dependency.

## Resolution Consistency

Resolution choices across the simulation should remain mutually coherent.

Material inconsistencies should be: identified, justified, assessed, and reflected in limitations where necessary.

## Resolution Change

Material changes in resolution should be governed.

A change may result from: model revision, new computational capacity, new data, changed scenario, new purpose, expanded scope, identified failure, or improved representation.

Resolution change may alter simulation outputs even where the underlying conceptual model remains unchanged.

## Resolution Change Impact

A material resolution change may affect:

Assumptions → Model & Environment → Inputs & Data → Scenario Design → Execution → Output Governance → Reality Correlation

RRM therefore treats resolution as a governed simulation dependency.

## Resolution Versioning

Where materially different resolutions are used, their configuration should remain distinguishable.

Simulation results produced at different resolution configurations should not automatically be treated as directly equivalent.

## Resolution Traceability

A governed resolution chain should support:

Simulation Purpose → Material Phenomenon → Resolution Object → Resolution Requirement → Resolution Configuration → Resolution Assessment → Resolution Gap → Resolution Status → Limitation

This allows later reconstruction of why a particular resolution was selected and what it could legitimately support.

## Relationship to Reality Selection

RSM establishes which parts of reality must be represented.

RRM cannot compensate for reality that was never selected.

Fine resolution of the wrong reality remains the wrong representation.

## Relationship to Reality Abstraction

RAM governs legitimate simplification.

RRM governs the granularity remaining after that simplification.

An abstraction may be legitimate conceptually while still being represented at insufficient resolution.

## Relationship to Representation Fidelity

RFM governs faithfulness.

RRM governs granularity.

These dimensions interact but must remain independently assessable.

## Relationship to Representation Limitation

RRM identifies limitations specifically arising from representation granularity.

RLM subsequently governs the complete limitation profile of the Reality Representation.

## Cross-Architecture Boundary

RRM does not govern actual Operational Reality.

That remains within ORA™.

RRM does not perform general validation.

That remains within VALIDOS®.

RRM does not govern general integrity.

That remains within INTEGROS®.

RRM governs specifically:

whether the internal representation of selected reality possesses sufficient granularity to preserve the material distinctions required by the simulation inquiry.

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## Minimum Implementation Framework

### 1. Identify Resolution Objects

Determine which represented phenomena, states, events, interactions, populations, dependencies, environments, and processes require explicit resolution governance. 2. Establish Resolution Requirements

Define the minimum granularity necessary for each material Resolution Object. 3. Configure Resolution Dimensions

Establish applicable spatial, temporal, state, behavioral, interaction, environmental, population, event, process, dependency, and failure resolution. 4. Assess Resolution Sufficiency

Determine whether the configured resolution preserves the material distinctions required by the simulation inquiry. 5. Identify Resolution Gaps

Record where the representation is too coarse to preserve a material phenomenon. 6. Assess Resolution Sensitivity

Determine whether plausible resolution changes materially affect simulation outcomes or interpretation. 7. Assess Cross-Component Resolution

Determine whether resolution differences across interacting components remain compatible. 8. Establish Resolution Status

Classify resolution as sufficient, conditional, partial, insufficient, or undetermined. 9. Establish Resolution Limitations

Bind unresolved resolution weaknesses to affected scenarios, conditions, outputs, and interpretations. 10. Preserve Resolution Traceability and Change Control

Maintain the complete resolution chain and reassess it after material change.

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## Governance Outputs

RRM may produce: Resolution Object Register, Resolution Requirement, Resolution Dimension Map, Spatial Resolution Record, Temporal Resolution Record, State Resolution Record, Behavioral Resolution Record, Interaction Resolution Record, Environmental Resolution Record, Population Resolution Record, Event Resolution Record, Process Resolution Record, Dependency Resolution Record, Failure Resolution Record, Measurement Resolution Record, Input Resolution Dependency Record, Output Resolution Boundary, Resolution Baseline, Resolution Threshold, Resolution Assessment, Resolution Sufficiency Record, Resolution Gap Record, Resolution Gap Classification, Resolution Trade-Off Record, Multi-Resolution Configuration, Adaptive Resolution Record, Adaptive Resolution Trigger, Cross-Component Resolution Assessment, Resolution Compatibility Record, Resolution Synchronization Record, Resolution Transformation Record, Aggregation Loss Record, Interpolation Dependency Record, Resolution Sensitivity Assessment, Resolution Convergence Record, Resolution Uncertainty Record, Conditional Resolution Record, Representation Resolution Status, Resolution Limitation Record, Resolution Change Record, Resolution Version Record, Resolution Traceability Map, and Representation Resolution Basis.

## Use Case 1 — Autonomous Vehicle Collision Simulation

### Scenario

An autonomous-vehicle simulation evaluates collision avoidance at urban intersections.

The vehicle dynamics and perception systems are represented accurately, but the simulation executes environmental changes only once per second.
Application

RRM determines that one-second Temporal Resolution cannot reliably represent: rapid pedestrian movement, braking initiation, sensor update cycles, vehicle response, and collision timing

required by the inquiry. Result

The representation receives:

## Resolution Insufficient — Collision Timing

even though other simulation components possess high fidelity.

The simulation must increase temporal resolution or restrict the legitimate interpretation of its collision-avoidance results. Use Case 2 — Spacecraft Micrometeoroid Simulation

### Scenario

A spacecraft simulation evaluates exposure to micrometeoroid impacts.

The spacecraft surface is represented using large spatial segments.
Application

RRM determines whether the current Spatial Resolution can preserve local differences in: shielding, component exposure, impact location, and structural vulnerability.

### Result

The existing resolution may remain sufficient for:

## Whole-Spacecraft Exposure Estimation

while being insufficient for:

Component-Level Damage Prediction.

The same representation therefore supports one inquiry but not the other. Use Case 3 — Industrial Workforce Simulation

### Scenario

A factory simulation represents all workers through one average productivity profile.

## The simulation is later used to examine emergency evacuation. Application

RRM identifies that population aggregation removes material differences involving: mobility, location, reaction time, role, shift distribution, and evacuation behavior.

### Result

The existing Population Resolution may remain sufficient for aggregate production forecasting but becomes:

## Resolution Insufficient — Emergency Evacuation Analysis

until the relevant population distinctions are represented.

## Architectural Position

Representation Resolution is the fourth internal governance space of the Reality Representation Standard (RRS).

The complete RRS progression is:

Reality Selection → Reality Abstraction → Representation Fidelity → Representation Resolution → Representation Limitation

RSM establishes:

What parts of reality must be represented?

RAM establishes:

How may their complexity be reduced?

RFM establishes:

How faithfully must the resulting representation correspond to reality?

RRM establishes:

At what granularity must the representation operate so that material phenomena remain distinguishable?

RLM next establishes:

Which limitations remain in the complete representation and how must they constrain subsequent simulation use and interpretation?

## Foundational Principle

A simulation has sufficient representation resolution only when its granularity preserves every distinction, transition, interaction, variation, and event that can materially affect the governed simulation inquiry.

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## Canonical Closing Statement

Representation Resolution Module establishes the governance conditions under which the granularity of simulated reality is selected, justified, assessed, bounded, changed, and traced. By governing spatial, temporal, state, behavioral, interaction, environmental, population, event, process, dependency, failure, and cross-component resolution while separating resolution from fidelity, numerical precision, and unnecessary detail, RRM ensures that simulation results cannot imply distinctions that the underlying representation was never capable of observing. Through resolution requirements, sufficiency assessment, gap identification, sensitivity analysis, multi-resolution governance, adaptive resolution, compatibility, synchronization, limitations, change control, and traceability, RRM ensures that simulation representation remains sufficiently granular for the purpose it is intended to serve without converting maximum detail into an unjustified substitute for methodological adequacy.

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# Canonical Source

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